MODULE – I COMPLETE ANSWERS (DBMS)

Based on the uploaded question bank.

SHORT ANSWER QUESTIONS (5 MARKS)

1. Explain Database Schema with Example

A **Database Schema** is the logical structure or blueprint of a database that defines how data is organized and how relationships are associated among them.

It specifies:

- Tables
- Attributes (Columns)
- Relationships
- Constraints

Example

Consider a Student Database.

Student Table

SID	Name	Dept
101	Anik	CSE

Schema:

STUDENT(

```
   SID INT PRIMARY KEY,
   Name VARCHAR(30),
   Dept VARCHAR(20)
)
```

The above structure is called the schema.

Characteristics

- Defined before data entry.
- Changes rarely.
- Describes database design.

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2. Define

(i) Database

A database is an organized collection of related data stored electronically for easy access, management, and updating.

Example

A college database storing:

- Student information
- Faculty details
- Course information

(ii) DBMS

A **Database Management System (DBMS)** is software used to create, maintain, and control access to databases.

Examples

- MySQL
- Oracle
- PostgreSQL
- SQL Server

Functions

- Data storage
- Data retrieval
- Security
- Backup and recovery
- Concurrency control

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3. Advantages of DBMS over File System

1. Reduced Data Redundancy

Same data need not be stored repeatedly.

2. Data Consistency

Updates are reflected everywhere.

3. Data Security

Unauthorized users can be restricted.

4. Data Sharing

Multiple users can access data simultaneously.

5. Backup and Recovery

Automatic backup and recovery mechanisms exist.

Additional Advantages

- Data Integrity
- Concurrency Control
- Centralized Management

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4. Explain the Role of Database Administrator (DBA)

A DBA is responsible for managing and controlling the database system.

Responsibilities

1. Database Design

Design tables and relationships.

2. Security Management

Grant and revoke privileges.

3. Backup and Recovery

Maintain backups.

4. Performance Tuning

Optimize queries and indexing.

5. User Management

Create and manage users.

6. Storage Management

Allocate storage efficiently.

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5. Explain Different Types of Database End Users

1. Casual Users

Access database occasionally.

Example

Manager checking reports.

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2. Naive Users

Use predefined applications.

Example

ATM users.

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3. Sophisticated Users

Write complex queries.

Example

Engineers and Analysts.

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4. Standalone Users

Use personal database software.

Example

MS Access users.

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5. Specialized Users

Develop specialized applications.

Example

AI researchers.

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6. Explain Database Instance with Example

A **Database Instance** is the actual data stored in the database at a particular moment.

Example

Schema:

STUDENT(SID, Name, Dept)

At Time T1:

SID	Name	Dept
101	Anik	CSE

At Time T2:

SID	Name	Dept
101	Anik	CSE
102	Rahul	EE

The data contents at T1 and T2 are different instances.

Difference

Schema	Instance
Structure	Actual Data
Rarely changes	Frequently changes

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7. Explain Logical Data Independence and Physical Data Independence

Data Independence means ability to modify schema without affecting higher levels.

Physical Data Independence

Changes in physical level do not affect logical level.

Example

- Change indexing
- Change file organization

Users remain unaffected.

Diagram

External Level

|

Conceptual Level

|

Physical Level

Changes occur here ↑

Logical Data Independence

Changes in conceptual level do not affect external views.

Example

Add new attribute:

Student(SID, Name, Dept)

to

Student(SID, Name, Dept, Phone)

Existing users continue working.

Comparison

Physical	Logical
Easier	Harder
Internal changes	Conceptual changes

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8. Purpose of Database System and Three Levels of Data Abstraction

Purpose

Database systems are designed to:

- Store large volumes of data

- Provide security
- Reduce redundancy
- Improve consistency
- Allow concurrent access

Three Levels of Data Abstraction

1. Physical Level

Describes how data is stored.

Example:

Files

Indexes

Blocks

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2. Logical Level

Describes what data is stored.

Example:

Student

Course

Faculty

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3. View Level

Describes what users see.

Example:

Teacher sees marks only.

Student sees own details only.

Diagram

View Level

|

Logical Level

|

Physical Level

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9. Different Types of Schema

1. External Schema

User-specific view.

Example:

Student View.

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2. Conceptual Schema

Overall logical structure.

Example:

Student, Faculty, Course relationships.

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3. Internal Schema

Physical storage details.

Example:

Indexes, blocks, file organization.

Diagram

External Schema

|

Conceptual Schema

|

Internal Schema

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10. Explain Database Design Process

Step 1: Requirement Collection

Gather user requirements.

Step 2: Conceptual Design

Create ER Diagram.

Step 3: Logical Design

Convert ER model to relational schema.

Step 4: Normalization

Remove redundancy.

Step 5: Physical Design

Choose:

- Indexing
- File organization

Step 6: Implementation

Create tables and constraints.

Step 7: Testing and Maintenance

Verify performance and security.

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11. Difference Between Metadata and Data Dictionary

Metadata

Data about data.

Example

Student.Name

Datatype = VARCHAR(30)

Data Dictionary

Repository storing metadata.

Contains:

- Table names
- Column names
- Constraints
- Data types

Difference

Metadata	Data Dictionary
Information about data	Collection of metadata
Small unit	Complete repository

Different Data Models

Hierarchical Model

Tree structure.

Network Model

Graph structure.

Relational Model

Tables.

Object-Oriented Model

Objects and classes.

Entity Relationship Model

Entities and relationships.

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12. Discuss Database Users and Administrators

Database Users

Naive Users

ATM customers.

Casual Users

Managers.

Sophisticated Users

Researchers.

Standalone Users

Personal database users.

Database Administrator

Responsible for:

- Security
- Backup
- Recovery
- Performance
- User management

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13. Explain Database Languages with Examples

1. DDL (Data Definition Language)

Used to define database structure.

Examples:

CREATE

ALTER

DROP

TRUNCATE

Example:

CREATE TABLE Student(

SID INT,

Name VARCHAR(30)

);

---

2. DML (Data Manipulation Language)

Manipulates data.

Examples:

INSERT

UPDATE

DELETE

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3. DQL (Data Query Language)

Retrieve data.

SELECT * FROM Student;

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4. DCL (Data Control Language)

Access control.

GRANT

REVOKE

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5. TCL (Transaction Control Language)

Transaction management.

COMMIT

ROLLBACK

SAVEPOINT

LONG ANSWER QUESTIONS (10 MARKS)

1. Discuss Three-Tier Database Architecture

Three-tier architecture separates database application into three layers.

1. Presentation Layer

User interface.

Examples:

- Browser
- Mobile App

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2. Application Layer

Business logic.

Examples:

- Java
- Python
- PHP

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3. Database Layer

Stores data.

Examples:

- MySQL
- Oracle

Diagram

```
+------------------+

| Presentation    |

| Layer           |

+------------------+

        |

+------------------+

| Application     |

| Layer           |

+------------------+

        |

+------------------+

| Database Layer  |

+------------------+
```

Advantages

- Security
- Scalability
- Easy Maintenance

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2. Physical Schema, Logical Schema and Conceptual Schema

Physical Schema

Describes storage details.

Examples:

- File organization
- Indexes

Logical Schema

Describes tables and attributes.

Example:

Student(SID, Name, Dept)

Conceptual Schema

Overall database structure.

Includes:

- Entities
- Relationships
- Constraints

Diagram

Users

|

External Schema

|

Conceptual Schema

|

Logical Schema

|

Physical Schema

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3. Architecture of Database System

Components

1. Users
2. Query Processor
3. DBMS Software
4. Database Storage

Diagram

Users

|

Query Processor

|

Database Manager

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Database Files

Functions

- Query execution
- Data retrieval
- Transaction processing

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4. Participation Constraints and Access Types in DML

Participation Constraints

Total Participation

Every entity must participate.

Example:

Every employee must belong to a department.

Representation:

Double Line

Partial Participation

Participation is optional.

Example:

Employee may manage a department.

Representation:

Single Line

Access Types in DML

Retrieval

SELECT

Insertion

INSERT

Modification

UPDATE

Deletion

DELETE

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5. Differentiate File Processing System and DBMS

Feature	File System	DBMS
Redundancy	High	Low
Security	Low	High
Backup	Difficult	Easy
Data Sharing	Limited	Excellent
Integrity	Poor	Strong
Concurrency	Poor	Good
Recovery	Difficult	Easy
Data Independence	No	Yes

Conclusion

DBMS is more efficient, secure, scalable, and reliable than traditional file systems.